STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-LIFT-EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2292-2300 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 2302-2351, the complete author proof (48 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_examples-defos.json (pb=2583, stmt 2292-2300, proof 2302-2351).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00DV: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2351–2351. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYC: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2351–2351. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DZ2: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2213–2234. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07M8: sources/stacks-examples-defos/pinned-original-source.tex, source lines 2351–2351. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A' \to A$ be a surjective map of Noetherian rings with nilpotent kernel. Let $A \to B$ be a finite type flat ring map. Let $\mathfrak{b} \subset B$ be an ideal such that $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is syntomic on the complement of $V(\mathfrak{b})$. Then B has a flat lift to A' if and only if the $\mathfrak{b}$-adic completion $B^\wedge$ has a flat lift to A'.*

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose an A-algebra surjection $P = A[x_1, \ldots, x_n] \to B$. Let $\mathfrak{p} \subset P$ be the inverse image of $\mathfrak{b}$. Set $P' = A'[x_1, \ldots, x_n]$ and denote $\mathfrak{p}' \subset P'$ the inverse image of $\mathfrak{p}$. (Of course $\mathfrak{p}$ and $\mathfrak{p}'$ do not designate prime ideals here.) We will denote $P^\wedge$ and $(P')^\wedge$ the respective completions.

Suppose $A' \to B'$ is a flat lift of $A \to B$, in other words, $A' \to B'$ is flat and there is an A-algebra isomorphism $B = B' \otimes_{A'} A$. Then we can choose an A'-algebra map $P' \to B'$ lifting the given surjection $P \to B$. By Nakayama's lemma (Algebra, Lemma 00DV) we find that B' is a quotient of P'. In particular, we find that we can endow B' with an A'-flat P'-module structure lifting B as an A-flat P-module. Conversely, if we can lift B to a P'-module M' flat over A', then M' is a cyclic module $M' \cong P'/J'$ (using Nakayama again) and setting $B' = P'/J'$ we find a flat lift of B as an algebra.

Set $C = B^\wedge$ and $\mathfrak{c} = \mathfrak{b}C$. Suppose that $A' \to C'$ is a flat lift of $A \to C$. Then C' is complete with respect to the inverse image $\mathfrak{c}'$ of $\mathfrak{c}$ (Algebra, Lemma 0DYC). We choose an A'-algebra map $P' \to C'$ lifting the A-algebra map $P \to C$. These maps pass through completions to give surjections $P^\wedge \to C$ and $(P')^\wedge \to C'$ (for the second again using Nakayama's lemma). In particular, we find that we can endow C' with an A'-flat $(P')^\wedge$-module structure lifting C as an A-flat $P^\wedge$-module. Conversely, if we can lift C to a $(P')^\wedge$-module N' flat over A', then N' is a cyclic module $N' \cong (P')^\wedge/\tilde{J}$ (using Nakayama again) and setting $C' = (P')^\wedge/\tilde{J}$ we find a flat lift of C as an algebra.

Observe that $P' \to (P')^\wedge$ is a flat ring map which induces an isomorphism $P'/\mathfrak{p}' = (P')^\wedge/\mathfrak{p}'(P')^\wedge$. We conclude that our lemma is a consequence of Lemma 0DZ2 provided we can show that B_g lifts to an A'-flat P'_g-module for $g \in \mathfrak{p}'$. However, the ring map $A \to B_g$ is syntomic and hence lifts to an A'-flat algebra B' by Smoothing Ring Maps, Proposition 07M8. Since $A' \to P'_g$ is smooth, we can lift $P_g \to B_g$ to a surjective map $P'_g \to B'$ as before and we get what we want. $\square$